Secure Data Ingestion & Analytics Platform on AWS

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1 Architecture Overview

The solution implements a secure, scalable AWS-based data lake architecture that supports:

- Batch ingestion from public sources (data.gov.sg)
- Private storage in Amazon S3
- Secure analytics using Athena
- Tableau integration via private connectivity

All traffic remains within AWS private networking after ingestion.

This architecture ensures:

- Secure public-to-private ingestion
- Private analytics via Athena PrivateLink
- No unnecessary public endpoints
- Strong encryption and monitoring
- High scalability and operational efficiency

2 Data Ingestion Architecture

Objective

Securely ingest large batch files (>100MB) from public endpoints into private AWS infrastructure.

Components

Layer	Services	Purpose
Scheduler	EventBridge	Triggers batch jobs
Orchestration	Step Functions	Controls workflow
Compute	Lambda / ECS (Private Subnet)	Handles API calls & large file downloads
Network	NAT Gateway	Secure outbound access to public API
Storage	Amazon S3 (Raw Zone)	Landing storage
Processing	AWS Glue	Data transformation
Final Storage	Amazon S3 (Processed Zone)	Analytics-ready data

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Security Controls

- Compute runs in **private subnets**
- No public access to S3
- NAT Gateway used only for outbound API calls
- S3 accessed via VPC Endpoint
- Encryption at rest using AWS KMS (AES-256)
- IAM roles enforce least privilege

3 Data Exploitation Architecture

Objective

Enable Tableau to query data using Athena securely while minimizing public exposure.

Components

Layer	Services	Purpose
Storage	Amazon S3 (Processed)	Data lake
Metadata	Glue Data Catalog	Schema management
Query	Amazon Athena	SQL query engine
BI	Tableau Server (EC2, Private Subnet)	Data visualization
Private Connectivity	Athena Interface VPC Endpoint (PrivateLink)	Secure connection

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Secure Query Flow

1. Tableau sends query using Athena Driver
2. Traffic routes through **Athena VPC Interface Endpoint (PrivateLink)**
3. Athena queries S3 via AWS internal network
4. Results return encrypted (TLS) to Tableau

No public internet exposure during query execution.

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4 Network Segmentation

The architecture uses **two logical zones**:

Ingestion VPC

- Handles public interaction
- Isolated from analytics environment

Exploitation VPC

- Hosts Tableau
- Connects privately to Athena via PrivateLink
- No inbound internet access

This segmentation prevents lateral movement in case of compromise.

5 Encryption Strategy

Encryption in Transit

- HTTPS (TLS/SSL)
- Applies to:
    - Tableau ↔ Athena
    - Lambda ↔ data.gov.sg

Encryption at Rest

- S3 encrypted using AWS KMS
- AES-256 standard

6 Scalability Considerations

- Lambda / ECS auto-scales
- Athena is serverless (no infrastructure management)
- S3 provides unlimited storage
- Glue supports distributed processing

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7 Performance Optimization

- Data partitioned by year/month
- Columnar format recommended (Parquet)
- Athena result caching enabled
- S3 lifecycle policies for cost optimization

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8 Assumptions

- Data ingestion is batch-based (not real-time)
- Files may exceed 100MB
- Tableau is hosted within AWS environment
- Dataset does not contain highly sensitive PII
- Government-grade encryption standards required

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9 Design Principles Applied

- Least privilege access
- Defense-in-depth
- Network isolation
- Serverless-first approach
- Managed services over self-managed infrastructure